matrix_a[0][1][7:0] matrix_a[1][1][7:0]

matrix_a[0][0][7:0] matrix_a[1][0][7:0]

matrix_b[0][1][7:0] matrix_b[1][1][7:0] matrix_b[0][0][7:0] matrix_b[1][0][7:0]

rst
start

clk +

index0_i RTL_ADD

I0[31:0] V=X"00000001" I1[31:0] O[31:0]
index_i RTL_MUX

S=32'sb00000000000000000000000000000001 I0[31:0] S=default I1[31:0] O[31:0] S[31:0]

index_i__0 RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] O[31:0] S[1:0]

index_i__1

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0] +

col0_i RTL_ADD

I0[31:0] V=X"00000001" I1[31:0] O[31:0]

index_reg[31:0]

RTL_REG_SYNC C

CED Q

RST

col_i RTL_MUX

S=32'sb00000000000000000000000000000001 I0[31:0] S=default I1[31:0] O[31:0] S[31:0]

col_i__1 RTL_ROM

A[31:0] O

col_i__0 RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] O[31:0] S[1:0]

col_i__2

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

row_i__0 RTL_ROM

A[31:0] O

col_reg[31:0]

RTL_REG_SYNC C

CED Q

RST

row_i__1 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0]

row_i__2 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0]+

row0_i RTL_ADD

I0[31:0] V=X"00000001" I1[31:0] O[31:0]

row_i__3

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

row_i RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] O[31:0] S[1:0]
row_reg[31:0]

RTL_REG_SYNC C

CED Q

RST state_i__0 RTL_ROM

A[31:0] O

matrix_a_i__0 RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0] matrix_a_i RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0]

matrix_b_i__0 RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0] matrix_b_i RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0]

state_i__1 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0]

matrix_a_i__1 RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0]

matrix_c_i__1 RTL_MUX S

S=1'b0 I0[16:0] S=1'b1 I1[16:0] O[16:0] matrix_c_i__6 RTL_MUX S

S=1'b0 I0[16:0] S=1'b1 I1[16:0] O[16:0]

matrix_b_i__1 RTL_MUX S

S=1'b0 I0[7:0] S=1'b1 I1[7:0] O[7:0]

state_i__2 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0] *RTL_MULT RTL_MULT

I0[8:0]

I1[8:0] O[16:0] matrix_c_reg[1][0]_i RTL_ROM
A[31:0] O matrix_c_reg[0][0]_i RTL_ROM
A[31:0] O matrix_c_i__7 RTL_MUX S

S=1'b0 I0[16:0] S=1'b1 I1[16:0] O[16:0]

matrix_c_reg[0][1]_i RTL_ROM

A[31:0] O matrix_c_reg[1][1]_i RTL_ROM

A[31:0] O

state_i__3

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]
state_i

RTL_MUX

V=B"01", S=2'b00 I0[1:0] V=B"10", S=2'b01 I1[1:0] S=2'b10 I2[1:0] S=default I3[1:0] O[1:0] S[1:0] +RTL_ADD

RTL_ADD

I0[16:0]

I1[16:0] O[16:0] matrix_c_reg[1][0]_i__0 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0]

matrix_c_reg[0][0]_i__0 RTL_MUX

S=32'sb00000000000000000000000000000000 I0

S=default I1 O

S[31:0] matrix_c_reg[0][1]_i__0 RTL_MUX

S=32'sb00000000000000000000000000000000 I0

S=default I1 O

S[31:0] matrix_c_reg[1][1]_i__0 RTL_MUX

S=32'sb00000000000000000000000000000001 I0

S=default I1 O

S[31:0]

state_reg[1:0]

RTL_REG_SYNC C

CED Q

RST

busy_i__0

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O

S[1:0] busy_i

RTL_MUX

S=2'b00 I0
S=2'b01 I1

S=2'b10 I2 O

S[1:0] matrix_c_i__0

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O

S[1:0] matrix_c_i RTL_MUX
S=2'b00 I0[16:0] S=2'b01 I1[16:0] O[16:0] S[1:0] matrix_c_i__2 RTL_MUX

S=2'b00 I0[16:0] S=2'b01 I1[16:0] O[16:0] S[1:0] matrix_c_i__3
RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O

S[1:0] matrix_c_i__4 RTL_MUX

S=2'b00 I0[16:0] S=2'b01 I1[16:0] O[16:0] S[1:0] matrix_c_i__5
RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O

S[1:0] matrix_c_i__8 RTL_MUX

S=2'b00 I0[16:0] S=2'b01 I1[16:0] O[16:0] S[1:0] matrix_c_i__9
RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

done_i

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

busy_reg

RTL_REG_SYNC C

CED Q

RST

matrix_c_reg[1][0][16:0]

RTL_REG_SYNC C

CED Q

RST

matrix_c_reg[0][0][16:0]

RTL_REG_SYNC C

CED Q

RST

matrix_c_reg[0][1][16:0]

RTL_REG_SYNC C

CED Q

RST

matrix_c_reg[1][1][16:0]

RTL_REG_SYNC C

CED Q

RST

done_reg RTL_REG_SYNC CD Q

RST

busy

matrix_c[1][0][16:0]

matrix_c[0][0][16:0]

matrix_c[0][1][16:0]

matrix_c[1][1][16:0]

done